Beyond Trajectory Statistics: Community-Consistent Human Mobility Generation via Evolving User–Location Graphs

Anonymous Author(s)

ABSTRACT

Web-based location services rely on human mobility data to model user-location interactions and support personalized services. However, collecting and sharing real trajectories is costly and often restricted by privacy concerns, motivating realistic synthetic mobility generation. Although recent generators incorporate collective mobility patterns, they primarily focus on individual trajectories and aggregate population and flow distributions, leaving evolving community structures formed by user-location interactions insufficiently modeled. To address this gap, we study community-consistent human mobility generation and propose MOTION, a Multi-agent framework for cOmmuniTy-consIstent human mObility geNeration. MOTION consists of **user agents**, **location agents**, and a global **manager agent**. MOTION detects communities from historical mobility using Leiden and extracts community-conditioned interaction priors to guide generation. Guided by these priors, user agents plan mobility, search for candidate destinations, and infer movements from the individual view, while selectively activated location agents infer likely arrivals from the community view. The manager coordinates both views, traces structural deviations to responsible interactions, and provides targeted reflection signals to revise user decisions, location attraction, and source-user selection. Experiments on two real-world datasets show that MOTION outperforms baselines by at least 4.05% across metrics for individual trajectories, population and flow distributions, and community structures. Further analyses demonstrate robustness, privacy preservation, and downstream utility. The code is available at https://anonymous.4open.science/r/MOTION-04C4.

CCS CONCEPTS

• **Information systems** → **Spatial-temporal systems**; **Data mining**.

KEYWORDS

Mobility Generation, Multi-Agent, Graph Generation, Community-Consistent.

1 INTRODUCTION

Web-based location services, including location-based social networks, point-of-interest (POI) recommendation, and online route planning [18, 39, 46], rely on human mobility data to understand how users interact with locations [1, 23, 38]. These interactions support personalized location recommendations and, when aggregated, provide valuable information for location demand analysis and other location-aware Web applications [10, 20, 21]. However, collecting and sharing user trajectories is often constrained by data scarcity, collection costs, and privacy concerns [2, 24, 37, 39]. Human mobility generation therefore provides realistic synthetic trajectories to support these services while reducing reliance on sharing sensitive individual mobility records [30, 37, 47].

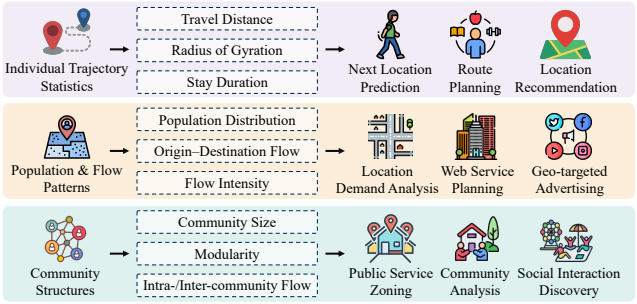


Figure 1: Three layers of human mobility patterns, their properties, and potential downstream applications.

To support Web applications, synthetic mobility should preserve three layers of mobility patterns [3, 8, 31, 40]. As illustrated in Fig. 1, at the individual layer, trajectory statistics characterize user mobility preferences, supporting personalized POI recommendation and route planning [11, 25]. At the population and flow layer, spatial distributions and origin-destination flows characterize aggregate mobility demand, supporting location demand analysis and geo-targeted Web services [20, 21]. At the community structure layer, shared destinations and recurring user-location interactions reveal mesoscopic community organization, providing a basis for community analytics, and social interaction discovery [13, 17, 43].

Existing studies have explored human mobility generation from both individual and collective perspectives. Mechanism-based methods [28] and deep learning models [6, 11] capture individual routines, transition preferences, and spatiotemporal dependencies. Large language model (LLM)-based methods [12, 15, 34, 36, 41] further model activity semantics and destination choices. At the collective level, d-EPR [27] exploits population-level location attractiveness, AgentMove [9] extracts shared mobility patterns across users, MobAgent [19] models heterogeneous population profiles, and CoDiffMob [47] and DeepMobility [44] incorporate population dynamics and collective flows into mobility generation. Despite these advances, existing methods primarily emphasize individual trajectory realism and aggregate population and flow patterns, leaving the evolving community structures formed by user-location interactions insufficiently modeled. Such aggregate patterns do not fully capture which users repeatedly visit shared locations, how these interactions organize communities, or how community relationships evolve over time [13, 17]. Therefore, mobility generation should go beyond matching trajectory statistics to explicitly preserve evolving community structures.

In this paper, we study community-consistent human mobility generation. To model community structures, we formulate mobility generation as dynamic user-location graph generation, where users and locations are represented as nodes and their interactions as edges. This unified representation explicitly captures the dependencies among users, locations, and mobility decisions, tracking user-location interactions and the resulting community structures

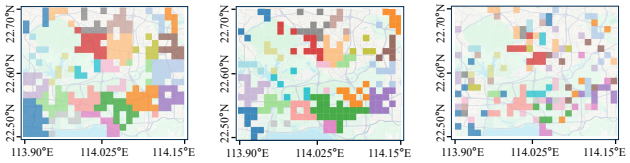
over time [26, 40]. Achieving this objective requires addressing the following three challenges:

• **How to represent dynamic community structures for mobility generation?** Community-consistent mobility generation requires priors that characterize how community structures and interaction patterns evolve with user-location interactions. These priors should preserve temporal and structural regularities without reproducing exact historical realizations. The first challenge is therefore to identify temporally coherent communities and extract representative interaction priors for generation.

• **How to generate user-location interactions efficiently for large-scale mobility datasets?** Under the graph-based formulation, each mobility decision creates an interaction between a user and a location, which should reflect both the user's mobility intention and the location's role. However, reasoning over all possible user-location pairs is computationally prohibitive, potentially requiring over one million inferences to generate a single day of hourly mobility for only 100 users. The second challenge is therefore to efficiently consider complementary user-side and location-side evidence.

• **How to align generated user-location graphs with realistic mobility structures?** Even when individual user-location interactions are plausible, their aggregation may still deviate from realistic community structures. Correcting such deviations requires identifying graph-level discrepancies, tracing them to the responsible user and location decisions, and determining how these decisions should be revised. The third challenge is therefore to translate structural deviations into targeted revisions of individual decisions.

To address these challenges, we propose MOTION, a Multi-agent framework for cOmmuniTy-consIstent human mObility geNeration. MOTION consists of **user agents** for target users, **location agents** for locations, and a global **manager agent**. Rather than learning previously unknown collective patterns through a trained predictor, MOTION extracts community-conditioned priors from historical mobility to guide generation and targeted reflection. Specifically, MOTION first constructs dynamic user-location graphs from historical trajectories and applies Leiden-based dynamic community detection to identify evolving communities. Within each community, MOTION employs metagraph-based temporal template matching to extract recurring mobility motifs, population and flow dynamics, and co-mobility patterns. These priors capture temporal and structural regularities without reproducing exact historical realizations, serving as guidance for generation and structural targets for reflection, addressing the **first** challenge. Guided by these priors, user agents and location agents generate user-location interactions from the individual and community views, respectively. Each user agent constructs a mobility plan based on mobility motifs, searches for candidate destinations, and infers movements using narrowed candidate-location contexts and reflection signals. In parallel, an active location selection mechanism activates relevant location agents based on population and flow deviations and moving users in nearby locations. Each activated location agent infers likely arriving users based on surrounding user contexts, population and flow dynamics, co-mobility patterns, and reflection signals. The two types of agents construct complementary individual-view and community-view graphs, addressing the **second** challenge. The manager agent coordinates these proposals based on their supporting evidence to construct an intermediate graph. It then identifies deviations from realistic community structures and population and flow patterns, traces them to responsible user and location decisions, and generates targeted reflection signals to revise user mobility plans, destination preferences, location attraction estimates, and source-user selections. Only affected agents regenerate their proposals, addressing the **third** challenge.



(a) Real Data (b) CoDiffMob (c) LLMob

Figure 2: Community structures detected from real and generated mobility. Different colors represent different communities on the same gridded map.

The main contributions can be summarized as follows:

• We study community-consistent human mobility generation and formulate it as dynamic user-location graph generation to explicitly model evolving community structures.

• We propose MOTION, a prior-guided multi-agent framework that coordinates user and location agents to generate individual- and community-view graphs, and employs manager-driven targeted reflection to preserve structural consistency.

• We conduct extensive experiments on two real-world datasets, demonstrating that MOTION outperforms baselines by at least 4.05% across the reported metrics. Additional analyses evaluate its robustness, privacy preservation, and downstream utility.

2 PRELIMINARY STUDY

2.1 Community Structural Consistency

A mobility community consists of users and locations connected through synchronized movements and interdependent population flows [17], capturing mesoscopic activity regions and interaction boundaries [13]. Although recent mobility generators incorporate collective mobility patterns, preserving community structures formed by user-location interactions remains challenging. To illustrate this limitation, we compare real mobility with CoDiffMob [47] and LLMob [36], which model collective mobility and use LLMs, respectively. We apply the same dynamic community detection procedure to all datasets and visualize the detected communities on the same gridded urban map. As shown in Fig. 2, real mobility exhibits large, spatially coherent communities. CoDiffMob recovers some coherent regions but produces more fragmented community assignments, whereas LLMob yields more scattered communities. These observations suggest that modeling individual and aggregate mobility patterns does not necessarily preserve community-level organization. Therefore, mobility generation should explicitly model and preserve evolving community structures.

2.2 Problem Definition

Given real-world mobility data and the initial mobility states of a set of target users, human mobility generation aims to generate realistic future mobility trajectories. Mathematically, let $\mathcal{U} = \{u_i\}_{i=1}^{N}$

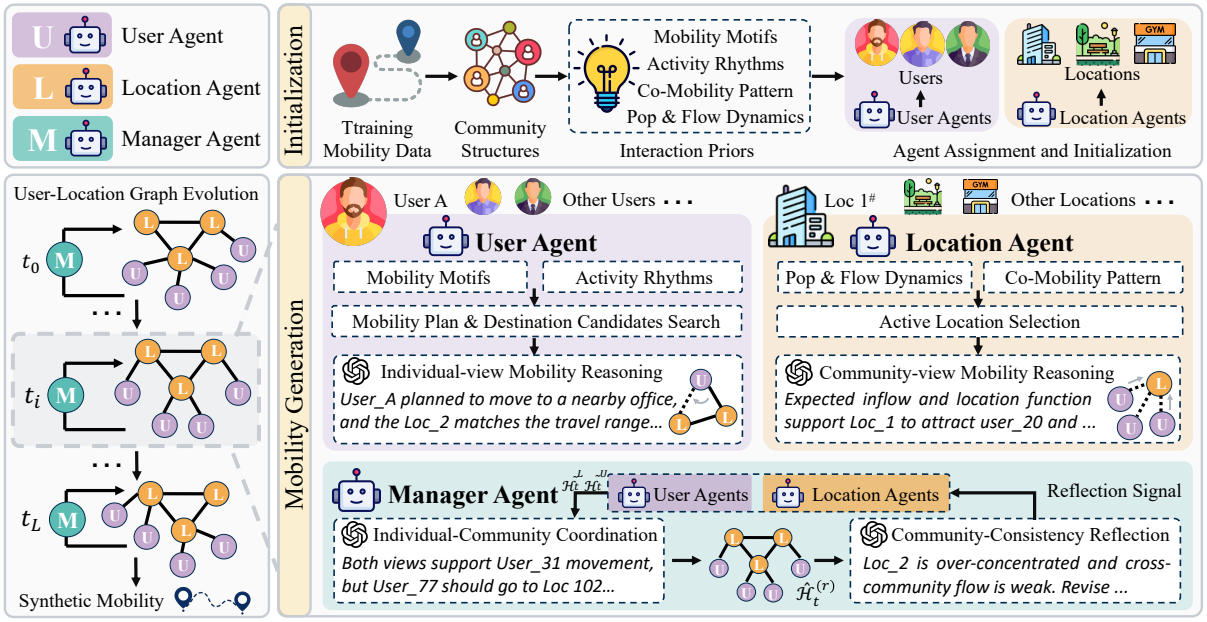


Figure 3: Overview of MOTION. MOTION extracts community structures and interaction priors from historical mobility to guide user- and location-agent generation from individual and community views. The manager coordinates their proposals and performs prior-guided targeted reflection to improve structural consistency.

and $\mathcal{L} = \{\ell_j\}_{j=1}^{M}$ denote the user and location sets. The historical mobility dataset is $\mathcal{S} = \{s_i\}_{i=1}^{N}$, where $s_i = [(\ell_0^i, t_0^i), \ldots, (\ell_{T_i}^i, t_{T_i}^i)]$ is the trajectory of user u_i. Given a set of target users $\hat{\mathcal{U}} = \{\hat{u}_i\}_{i=1}^{N^*}$ and their initial states $\mathcal{X}_0 = \{(\ell_0^i, t_0^i)\}_{i=1}^{N^*}$, let $\mathcal{G}$ denote a mobility generation model. The generated trajectories are formulated as:

$$\hat{\mathcal{S}} = \{\hat{s}_i\}_{i=1}^{N^*} = \mathcal{G}(\mathcal{S}, \mathcal{X}_0). \tag{1}$$

3 METHODOLOGY

3.1 Framework Overview

The overall framework of MOTION is illustrated in Fig. 3. MOTION consists of a **user agent** for each target user, a **location agent** for each location, and a global **manager agent**. During initialization, MOTION constructs dynamic user-location graphs from historical trajectories, detects coherent communities using temporally smoothed Leiden-based community detection, and extracts community-conditioned interaction priors using metagraph-based mining. Rather than training a predictor to discover new collective patterns, MOTION uses these priors to guide agent decisions and provide structural reference targets for reflection. During generation, user agents construct mobility plans, search for candidate destinations, and infer movements from the individual view. Meanwhile, an active location selection mechanism activates structurally relevant location agents to infer likely user arrivals from the community view. The manager coordinates these complementary proposals, identifies structural deviations from the extracted priors, and traces them to responsible user and location decisions. It then provides targeted reflection signals for the affected agents to revise their proposals, and the refined graph conditions the next time step.

3.2 Mobility Prior Initialization

To represent the interaction and community structures emerging across users, MOTION models historical mobility as dynamic user-location graphs, and then extracts temporally coherent communities and community-conditioned interaction priors.

3.2.1 Dynamic User-Location Graph Construction. MOTION represents human mobility as a sequence of dynamic user-location graph snapshots. Each snapshot contains time-varying user-location edges indicating users' current locations and static location-location edges capturing geographic proximity and historical mobility connectivity. Specifically, let $\mathcal{U}$ and $\mathcal{L}$ denote the historical user and location sets. After discretizing trajectories into T time intervals, MOTION constructs:

$$\mathbb{H} = \{\mathcal{H}_1, \ldots, \mathcal{H}_T\}, \qquad \mathcal{H}_t = \left(\mathcal{U} \cup \mathcal{L}, \mathcal{E}_t^{UL} \cup \mathcal{E}^{LL}\right), \tag{2}$$

where $(u_i, \ell_j) \in \mathcal{E}_t^{UL}$ indicates that user u_i is located at ℓ_j at time t. The static edge set $\mathcal{E}^{LL}$ connects locations according to geographic proximity and aggregate historical flows, providing a spatial backbone shared across snapshots. Changes from $\mathcal{E}_{t-1}^{UL}$ to $\mathcal{E}_t^{UL}$ encode individual movements and jointly determine location populations and inter-location flows. Therefore, human mobility is represented as sequential user-location graphs rather than isolated location transitions.

3.2.2 Community-Conditioned Interaction Prior Mining. To capture community structures and interaction patterns, MOTION employs temporally smoothed Leiden-based community detection. Let C_t denote the community partition of $\mathcal{H}_t$. For $t > 1$, Leiden optimizes the Constant Potts Model (CPM) objective [35], initialized with the preceding partition C_{t-1}^*. Temporal continuity is further encouraged

by stochastically retaining each location's previous community assignment, while allowing memberships to evolve with user-location interactions. Each user is then assigned to the community of its current location. Based on the detected communities, MOTION employs metagraph-based temporal template matching to extract community-conditioned interaction priors. Individual-level priors include mobility motifs, motif transitions, and activity rhythms, while community-level priors capture community memberships and sizes, population and flow dynamics, and co-mobility patterns. Together, these priors characterize how individual movements shape evolving community structures and guide subsequent generation. Representative examples are provided in Appendix E.

3.3 User Agent

MOTION assigns a user agent to each target user. Each agent maintains the user's mobility state, constructs a daily mobility plan, searches for candidate destinations, and infers the next location from the individual view.

3.3.1 Individual Mobility Planning. To organize daily mobility decisions, the user agent first samples a community for user $\hat{u}_i$ from the affiliation distribution $p(c \mid \ell_0^i)$ extracted from historical mobility. The user agent then samples a sequence of mobility motifs conditioned on the assigned community, time of day, and motif transition patterns. The transition between consecutive motifs is modeled as:

$$m_{r+1}^i \sim p\left(m_{r+1} \mid m_r^i, c_i, h_r\right), \tag{3}$$

where m_r^i denotes the current motif, c_i denotes the assigned community, and h_r denotes its activation time. Each motif is instantiated as a sequence of stay and move actions based on extracted activity rhythms, destination semantics, travel-distance ranges, and return patterns. The resulting plan specifies movement timing, destination intentions, and the expected movement structure. Users with compatible plans and community memberships are further associated through co-mobility patterns to support location-side reasoning. During generation, targeted reflection may adjust planned actions, execution times, travel ranges, or destination intentions.

3.3.2 Individual-View Mobility Reasoning. At each time step, the user agent executes the active segment of its mobility plan to determine the next location. For a stay action, the current user-location edge is retained. For a return action, the agent revisits a recorded location to preserve the mobility motif. To reason efficiently and reliably about other movements, MOTION searches for candidate destinations within the planned travel-distance range and evaluates their semantic compatibility, geographic distance, regional attraction, and community flow evidence. The user agent jointly interprets these signals together with reflection signals to select the next location. The decisions of all user agents are aggregated into the individual-view graph $\widetilde{\mathcal{H}}_t^U$.

3.4 Location Agent

MOTION assigns a location agent to each location to model mobility from the receiver-side community view. An active location selection mechanism selectively activates relevant location agents

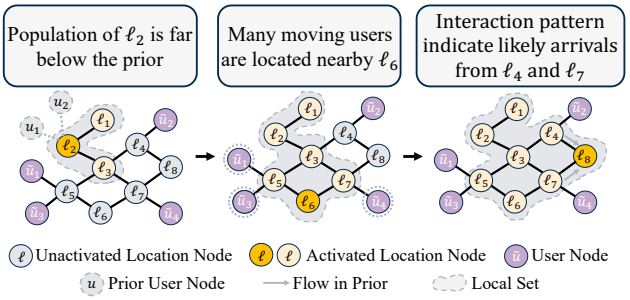


Figure 4: Local-set formation. Locations and their neighborhoods are activated according to population dynamics, nearby moving users, and prior-supported incoming flows and are then merged into a local set.

to infer likely user arrivals. This receiver-side reasoning complements user agents' destination decisions and promotes realistic collective interaction patterns.

3.4.1 Active Location Selection. To improve reasoning efficiency, MOTION identifies active locations based on deviations between current and expected populations, moving users in nearby or graph-connected source locations, and prior-supported incoming flows and co-mobility patterns. MOTION then groups these locations and their neighbors into local sets, as illustrated in Fig. 4. From each local set, MOTION randomly samples at most `active_m` locations for agent activation. The sampled locations are merged into the final active location set. This strategy ensures coverage across local mobility regions while preventing excessive activation of popular locations.

3.4.2 Community-View Mobility Reasoning. At each time step, each activated location agent identifies candidate users from graph-connected source locations and infers their potential arrivals. The location agent considers users' current locations, mobility plans, activity intentions, community memberships, and co-mobility relations, together with the target location's semantic role, expected population, and incoming flows. It then selects users whose arrivals align with plausible graph connections, activity intentions, and community-conditioned interaction patterns. Co-mobility evidence further supports synchronized movements toward shared destinations. If multiple locations select the same user, MOTION retains the proposal best supported by population, flow, and co-mobility priors. All location-agent proposals are aggregated into the community-view graph $\widetilde{\mathcal{H}}_t^L$ for subsequent coordination with the individual-view graph by the manager agent.

3.5 Manager Agent

The manager agent coordinates individual- and community-view graphs and uses extracted interaction priors to diagnose structural deviations and guide targeted reflection. This prior-guided process aims to preserve individual mobility intentions while improving consistency with realistic community structures.

3.5.1 Individual-Community Coordination. At each time step, the manager agent coordinates the individual-view graph $\widetilde{\mathcal{H}}_t^U$ and the community-view graph $\widetilde{\mathcal{H}}_t^L$. For each user, MOTION considers the

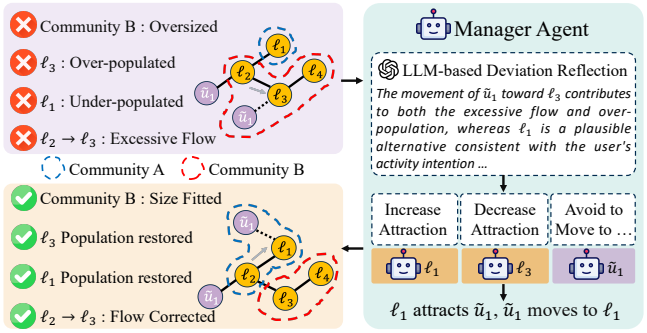


Figure 5: Community-consistency reflection. MOTION bridges generated and real communities, diagnoses deviations, attributes them to related decisions, and sends reflection signals to regenerate the affected proposals.

user agent's mobility plan, movement history, intended activity, proposed destination, and rationale, together with the location agent's attraction estimates, expected population and flows, community role, co-mobility evidence, and rationale. When both views agree, their proposal is retained. Otherwise, the manager uses LLM reasoning to evaluate the supporting evidence and resolve conflicts by balancing individual intentions with community-conditioned interaction priors. The resolved interactions form an intermediate graph $\widehat{\mathcal{H}}_t^{(r)}$ at coordination round r, which is subsequently evaluated for structural consistency.

3.5.2 Community-Consistency Reflection. Even individually plausible interactions may collectively deviate from realistic population, flow, and community structures. MOTION therefore iteratively coordinates agent proposals and refines the generated graph using historical interaction priors:

$$\begin{aligned} \widehat{\mathcal{H}}_t^{(r)} &= \text{Coord}\left(\widetilde{\mathcal{H}}_t^{U,(r)}, \widetilde{\mathcal{H}}_t^{L,(r)}\right), \\ \widehat{\mathcal{H}}_t^{(r+1)} &= \text{Reflect}\left(\widehat{\mathcal{H}}_t^{(r)}, \mathcal{P}_t\right), \end{aligned} \tag{4}$$

where $\mathcal{P}_t$ denotes community-conditioned priors extracted from historical mobility, and Reflect represents prior-guided diagnosis, targeted agent regeneration, and re-coordination. To identify structural deviations, MOTION applies the same temporally smoothed community detection procedure to the intermediate graph and aligns generated communities with time-corresponding historical reference communities using location-node Jaccard similarity. It then compares community memberships and sizes, location populations, and intra- and inter-community flows against the extracted priors. As illustrated in Fig. 5, MOTION identifies locations with mismatched community memberships or substantial population and flow deviations. For each deviating location, the manager examines users staying at, departing from, or arriving at that location, along with flow-connected source and destination locations, to identify the responsible interaction decisions. Based on this diagnosis, the manager provides targeted reflection signals to the corresponding agents. User agents may revise planned actions, execution times, travel ranges, or destination intentions, while location agents may adjust attraction estimates, preferred source locations, or candidate users. Only the affected agents regenerate their proposals, which are then re-coordinated to obtain $\widehat{\mathcal{H}}_t^{(r+1)}$.

Table 1: Evaluation metrics for the three pattern layers.

Pattern Layer	Evaluation Metrics
Individual Trajectory Statistics	Distance, Radius, LocNum
Population and Flow Distributions	Pop-EMD, Flow-EMD, G-Rank
Community and Graph-Structural Consistency	Motif, Comm-Size, Comm-Mem, Comm-Trans

4 EXPERIMENTS

4.1 Experimental Setup

4.1.1 Datasets and Experimental Settings. We evaluate MOTION on two real-world datasets. The **private car dataset** [22] contains two weeks of GPS traces from 2,342 private vehicles, while the **mobile phone dataset** [34] contains two weeks of GPS traces from 5,352 users. Both datasets were collected with explicit authorization from users and data providers and were de-identified. We adopt an 8:2 user-level train-test split and reserve the final day for evaluation. Train and test users are disjoint, and test users' historical trajectories are excluded from prior extraction or training. We employ GPT-4o as the default LLM backbone for MOTION and baselines. Dataset statistics, ethical information, and POI profiles are provided in the supplementary material; detailed settings appear in Appendix C.

4.1.2 Baselines and Metrics. We compare MOTION with seven representative mobility generators, including the GAN-based model MoveSim [11], the collaborative generator DeepMobility [44], the diffusion-based model CoDiffMob [47], LLM-based methods LLMob [36] and LLM-COD [41], and the agent-based models TrajLLM [14] and AgentMove [9]. Detailed descriptions of the baselines are provided in Appendix B. We evaluate MOTION at the three pattern layers summarized in Table 1. Detailed metric definitions are provided in Appendix D.

4.2 Overall Performance

Table 2 reports the overall performance on the private car and mobile phone datasets. MOTION achieves the best results across all metrics on both datasets, demonstrating its effectiveness in preserving individual, population, and community-level mobility patterns. At the **individual trajectory statistics** level, MOTION reduces Distance by 11.36% over LLMob on the private car dataset and by 12.87% over DeepMobility on the mobile phone dataset. These improvements suggest that community-conditioned priors and user-agent reasoning jointly support realistic individual movements. At the **population and flow distributions** level, MOTION reduces Flow-EMD by 4.05% over CoDiffMob on the private car dataset and by 16.83% over TrajLLM on the mobile phone dataset, highlighting its ability to capture collective dynamics. At the **community and graph-structural consistency** level, MOTION improves Motif, Comm-Size, and Comm-Mem by 14.27%, 6.95%, and 7.47% on the private car dataset, and Motif and Comm-Mem by 11.38% and 8.04% on the mobile phone dataset, respectively. It also achieves the best Comm-Trans scores of 0.3190 and 0.4016, demonstrating superior community evolution consistency. Overall, these results demonstrate that MOTION preserves community organization and community structure temporal evolution while maintaining realistic individual mobility patterns.

Table 2: Overall performance on the private car and mobile phone datasets.

Methods	Individual Trajectory Statistics			Population and Flow Distributions			Community and Graph-Structural Consistency			
	Distance↓	Radius↓	LocNum↓	Pop-EMD↓	Flow-EMD↓	G-Rank↓	Motif↓	Comm-Size↓	Comm-Mem↑	Comm-Trans↑
Private Car Dataset										
MoveSim	0.2257±0.0034	0.1781±0.0029	0.2575±0.0038	0.1037±0.0020	0.1014±0.0019	0.3475±0.0051	0.2918±0.0044	0.2203±0.0035	0.2756±0.0033	0.2334±0.0007
DeepMobility	0.2086±0.0053	0.1185±0.0019	0.2190±0.0032	0.0565±0.0018	0.0555±0.0024	0.2614±0.0030	0.1563±0.0038	0.0977±0.0012	0.5335±0.0041	0.2707±0.0004
CoDiffMob	0.2106±0.0035	0.1176±0.0016	0.2494±0.0034	0.0422±0.0009	0.0420±0.0011	0.2851±0.0042	0.2180±0.0017	0.0676±0.0010	0.5365±0.0022	0.2773±0.0009
LLMob	0.1928±0.0018	0.1175±0.0017	0.1463±0.0018	0.0428±0.0005	0.0458±0.0009	0.2558±0.0037	0.1920±0.0030	0.0723±0.0010	0.4995±0.0041	0.2596±0.0004
LLM-COD	0.2374±0.0029	0.2232±0.0034	0.1976±0.0030	0.0759±0.0015	0.0729±0.0015	0.2259±0.0034	0.1919±0.0029	0.0754±0.0013	0.4684±0.0055	0.2682±0.0005
TrajLLM	0.2364±0.0031	0.1880±0.0030	0.2207±0.0033	0.0797±0.0016	0.0767±0.0017	0.2261±0.0032	0.2084±0.0032	0.0939±0.0017	0.3750±0.0018	0.2389±0.0008
AgentMove	0.2611±0.0037	0.2257±0.0038	0.2479±0.0035	0.0436±0.0009	0.1753±0.0028	0.2233±0.0021	0.2933±0.0045	0.0680±0.0009	0.5194±0.0029	0.2369±0.0040
MOTION	**0.1709**±0.0015	**0.0958**±0.0014	**0.1258**±0.0015	**0.0402**±0.0004	**0.0403**±0.0007	**0.2079**±0.0018	**0.1340**±0.0018	**0.0629**±0.0007	**0.5766**±0.0012	**0.3190**±0.0005
Improvement	11.36%	18.47%	14.01%	4.74%	4.05%	6.90%	14.27%	6.95%	7.47%	15.04%
Mobile Phone Dataset										
MoveSim	0.2382±0.0035	0.1920±0.0032	0.3151±0.0045	0.1804±0.0030	0.1777±0.0031	0.3018±0.0047	0.3387±0.0050	0.2231±0.0042	0.4366±0.0050	0.2253±0.0034
DeepMobility	0.1569±0.0018	0.1426±0.0019	0.1996±0.0026	0.1278±0.0014	0.1264±0.0056	0.3505±0.0059	0.2892±0.0034	0.1855±0.0027	0.4550±0.0029	0.2622±0.0030
CoDiffMob	0.2635±0.0039	0.1496±0.0027	0.1911±0.0025	0.1165±0.0013	0.1163±0.0013	0.3478±0.0037	0.2669±0.0051	0.1320±0.0041	0.4591±0.0067	0.2531±0.0025
LLMob	0.1778±0.0029	0.1834±0.0030	0.1116±0.0018	0.1277±0.0025	0.1274±0.0026	0.3544±0.0052	0.2803±0.0043	0.2143±0.0045	0.4601±0.0062	0.3249±0.0032
LLM-COD	0.1598±0.0026	0.1512±0.0025	0.2007±0.0031	0.1139±0.0022	0.1134±0.0022	0.2982±0.0043	0.2904±0.0045	0.1504±0.0032	0.4593±0.0041	0.2803±0.0036
TrajLLM	0.1917±0.0031	0.1955±0.0033	0.1852±0.0030	0.1105±0.0020	0.1105±0.0020	0.2960±0.0028	0.2691±0.0038	0.2317±0.0048	0.4517±0.0049	0.3574±0.0049
AgentMove	0.2257±0.0036	0.1443±0.0022	0.2818±0.0040	0.1297±0.0026	0.1295±0.0027	0.3508±0.0051	0.2426±0.0032	0.1921±0.0040	0.4367±0.0062	0.3270±0.0044
MOTION	**0.1367**±0.0020	**0.1263**±0.0018	**0.0955**±0.0014	**0.0925**±0.0015	**0.0919**±0.0015	**0.2773**±0.0022	**0.2150**±0.0024	**0.1170**±0.0025	**0.4971**±0.0025	**0.4016**±0.0026
Improvement	12.87%	11.43%	14.43%	16.29%	16.83%	6.32%	11.38%	11.36%	8.04%	12.37%

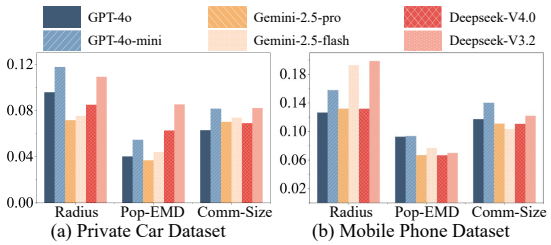
Table 3: Ablation results on both datasets.

Variants	Dist.↓	Radius↓	P-EMD↓	C-Size↓	C-Mem↑	C-Trans.↑
Private Car Dataset						
W/o User	0.1939	0.1918	0.0459	0.0887	0.4868	0.1590
W/o Location	0.1817	0.1050	0.0524	0.0881	0.5320	0.1903
W/o Manager	0.2054	0.1331	0.0486	0.0831	0.4677	0.1405
W/o Reflection	0.1949	0.1436	0.0556	0.1078	0.4855	0.1604
W/o Community	0.2178	0.1544	0.0616	0.0991	0.4615	0.1287
Single-Agent	0.2367	0.1174	0.0548	0.0792	0.5247	0.1339
Deterministic	0.2511	0.1219	0.0665	0.0841	0.4925	0.0942
MOTION	**0.1709**	**0.0958**	**0.0402**	**0.0629**	**0.5766**	**0.1990**
Mobile Phone Dataset						
W/o User	0.1767	0.2367	0.1025	0.1608	0.4358	0.3005
W/o Location	0.1678	0.1674	0.1657	0.1975	0.4132	0.2772
W/o Manager	0.1863	0.1763	0.1327	0.1462	0.4682	0.2442
W/o Reflection	0.1889	0.1744	0.1967	0.1959	0.4173	0.2618
W/o Community	0.1927	0.1901	0.1419	0.2400	0.2997	0.2245
Single-Agent	0.1848	0.1769	0.1246	0.1735	0.4666	0.3131
Deterministic	0.2201	0.1930	0.1903	0.1368	0.4771	0.2987
MOTION	**0.1367**	**0.1263**	**0.0925**	**0.1170**	**0.4971**	**0.3819**

Dist.=Distance; P-EMD=Pop-EMD; C-Size/Mem/Trans=Comm-Size/Mem/Trans.

4.3 Ablation Analysis

We conduct ablation studies to examine the contributions of community priors, agent roles, coordination, and reflection. **W/o User** and **W/o Location** remove the corresponding agents, while **W/o Manager** replaces manager coordination with fixed-weight proposal fusion and disables reflection. **W/o Community** removes community priors, while **W/o Reflection** retains manager coordination but disables reflection. **Single-Agent** uses the same priors, candidate sets, and reflection objectives as MOTION, but jointly considers user- and location-side evidence within a single agent without role separation. **Deterministic** retains the same information and reflection mechanism but replaces LLM-based reasoning with fixed-weight decision rules, whose parameters are optimized on a held-out validation subset of the training data. As shown in Table 3, MOTION achieves the best results across all six metrics on both datasets. Removing the user agent degrades mobility and structural consistency, while removing the location agent yields the largest Pop-EMD among role-removal variants. Removing reflection reduces Comm-Trans from 0.1990 to 0.1604 and from 0.3819 to 0.2618 on the two datasets, while removing the manager further reduces it to 0.1405 and 0.2442, respectively. Removing community priors increases Comm-Size from 0.1170 to 0.2400 and reduces Comm-Mem from 0.4971 to 0.2997 and Comm-Trans from 0.3819 to 0.2245 on the mobile phone dataset. Moreover, MOTION outperforms both same-information controls, supporting the benefits of role-specific reasoning and coordination beyond shared priors. Appendix **??** further compares MOTION with baselines supplied with reflection information.

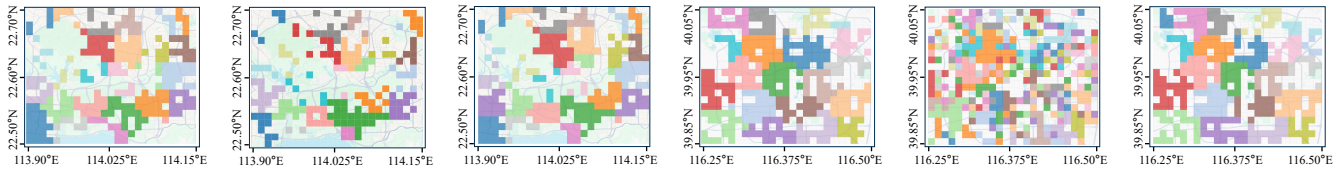


(a) Private Car Dataset (b) Mobile Phone Dataset

Figure 6: Impact of different LLM backbones on the private car and mobile phone datasets.

4.4 LLM Backbone Analysis

We evaluate the impact of different LLM backbones by replacing the default GPT-4o with alternative models. As shown in Fig. 6, backbone choice affects both individual mobility realism and community consistency, highlighting the importance of semantic understanding and coordination reasoning. MOTION remains effective across different backbones. We adopt GPT-4o for the main experiments based on its overall generation quality, although other models achieve better results on certain metrics.



(a) Real (b) CoDiffMob (c) MOTION (d) Real (e) LLMob (f) MOTION

Figure 7: Community structures detected from real and generated mobility. (a)–(c) Private car dataset; (d)–(f) Mobile phone dataset. Different colors represent different communities on the same gridded map.

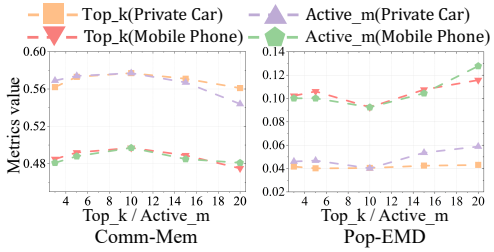


Figure 8: Sensitivity to `top_k` and `active_m` on the private car and mobile phone datasets.

4.5 Community Preservation Analysis

To evaluate community structure preservation, we apply the same community detection procedure to real and generated user-location graphs and visualize the resulting communities in Fig. 7. CoDiffMob produces fragmented communities, while LLMob exhibits more mixed community assignments. In contrast, MOTION generates spatially coherent communities whose geographic extents and boundaries more closely resemble those of real mobility, demonstrating better preservation of mesoscopic community structures. Appendix F further shows that MOTION achieves the highest NMI under randomized, UL-only and UL-shuffled settings, suggesting robustness to backbone perturbations.

4.6 Hyperparameter Analysis

We analyze two key hyperparameters: `top_k`, which determines the number of destination candidates retrieved for each user agent, and `active_m`, which limits the number of locations sampled from each local set for agent activation. As shown in Fig. 8, Comm-Mem and Pop-EMD generally improve as both parameters increase to 10 but deteriorate beyond this point, suggesting that excessive candidates introduce less relevant evidence and reasoning noise. These results highlight the importance of balancing evidence coverage with focused agent reasoning.

4.7 Cross-Dataset Prior Recombination

We evaluate cross-dataset prior recombination by transferring individual mobility motifs from a source dataset while using target-specific population, flow, co-mobility, and community priors to guide generation. This setting evaluates prior recombination rather than target-prior-free cross-city generalization. As shown in Fig. 12 in Appendix H, Distance and Radius show larger discrepancies due to differences in mobility characteristics and spatial scales, while population-flow and community-level metrics remain relatively stable. These results suggest that target-specific priors help preserve structural consistency across datasets.

Table 4: Performance on multiple held-out days (mean ± standard deviation). Bold indicates the best result.

Methods	Distance↓	Pop-EMD↓	Comm-Size↓	Comm-Mem↑
DeepMobility	0.217 ± 0.011	0.053 ± 0.004	0.097 ± 0.006	0.536 ± 0.004
CoDiffMob	0.220 ± 0.009	0.043 ± 0.003	0.070 ± 0.001	0.549 ± 0.003
LLMob	0.277 ± 0.013	0.057 ± 0.007	0.143 ± 0.002	0.449 ± 0.049
MOTION	**0.164** ± 0.006	**0.038** ± 0.002	**0.064** ± 0.003	**0.575** ± 0.010

Table 5: End-to-end token consumption and latency per generated user-day, and component costs per reasoning call.

Method	End-to-End / User-Day				MOTION Components / Call				
	Private Car		Mobile Phone		Component	Private Car		Mobile Phone	
	Tokens	Lat.	Tokens	Lat.		Tokens	Lat.	Tokens	Lat.
TrajLLM	43,804	231	44,185	230	User	436.1	1.57	372.3	2.21
LLMob	**40,647**	255	45,023	230	Location	1,345.0	2.15	1,462.0	1.98
AgentMove	42,331	264	48,159	244	Coordination	720.5	2.07	806.3	2.06
MOTION	40,662	**229**	**43,132**	**224**	Reflection	866.4	3.77	981.0	3.11

Lat.=latency in seconds; User and Location denote agents.

4.8 Temporal Robustness Analysis

We evaluate temporal robustness across multiple held-out days and longer generation horizons. As shown in Table 4, MOTION achieves the best results across Distance, Pop-EMD, Comm-Size, and Comm-Mem, demonstrating consistent performance across test days. As shown in Table 10 in Appendix J, we further compare daily resetting with continuous generation over two to four days. Although daily resetting performs better, continuous generation maintains relatively stable population distributions and community consistency over longer horizons.

4.9 Complexity Analysis

Table 5 reports user-day generation costs and component-level costs. MOTION achieves lower latency than LLMob and AgentMove with comparable or fewer tokens. We further conduct scaling experiments and find that costs scale approximately linearly with user count but remain relatively stable across location counts and reflection rounds. Active location selection costs $O(U_h + L_h \log L_h)$ per step, while local-set formation requires $O(d_{\text{in}} + 1)$ per active location. Potential optimizations include parallel execution and indexed retrieval.

4.10 Case Study

Fig. 9 illustrates a graph update where user and location agents propose movements based on individual intentions and community priors, respectively. The manager coordinates both views, resolves conflicts, and provides targeted reflection signals to correct structural deviations. Detailed prompts are provided in Appendix K.

Example of Multi-Agent Reasoning, Coordination and Reflection

User Agent 418: User 418 at 8:00 plans to MOVE to a Science & Education & Culture Area within 1km~1.5km). Candidate Locations are [loc 298, Shopping & Consumer Goods, dist=1.11km, gravity=12.660 flow_out=17.9; loc 317 Science & Education & Culture dist=1.11km gravity=6.242 flow_out=9.1, …]. Agent Decision: {"next_location_id": 317, "reason": "Best match for 'Science & Education & Culture' with good gravity and flow_out, and ideal distance."}**User Agent 452:** …

Location Agent 317: Loc 317 is a Science & Education & Culture activity center with 12 users here now. Scaled history population and flow at 8:00 are 12.9 and 16.62. Surrounding Locations with moving users are [Loc 209, Hotels & Accommodations area, moving_user_num: 29, hist_flow_in: 22.46; Loc 306, …]. Surrounding MOVING Users are [User 418, Daily Plan is…, co-mob users are…, now is in Loc 209, moving to a Science & Education & Culture Area; ….]. Agent Decision: {"users_to_attract": ["418","220"], "reason": "Population is under-populated, so only attract strong matches. Prioritize users from Hotels & Accommodations (loc 209)."}**Location Agent 298:** …

Manager Agent: Scaled history movement at 8:00 is13.8~25.7. [User 418, User Agent Decision and reason are …, Location Agent Decision and reason are…., User 258, …]. Agent Decision is {{"user_id": 418, "resolved_to_loc": 317,"action": "MOVE","reason": "Both agents agree on MOVE to 317; loc 317 is under-populated."}{"user_id": 901,"resolved_to_loc": 107,"action": "STAY","reason": "User agent proposes STAY; no strong real-data signal to override."},…}. Reflections: {"location_signals": [{"loc_id": 106,"signal_type": "decrease_attraction","detail": "Location 106 (Hotels & Accommodations) is over-populated; reduce attraction."},…], [{"user_id": "183","signal_type": "modify_plan_action","detail": "Switch planned action to STAY at Shopping & Consumer Good to reduce overpopulation."},…]}.

Figure 9: An example of multi-agent reasoning. Colored regions highlight the key inputs and outputs of each agent role.

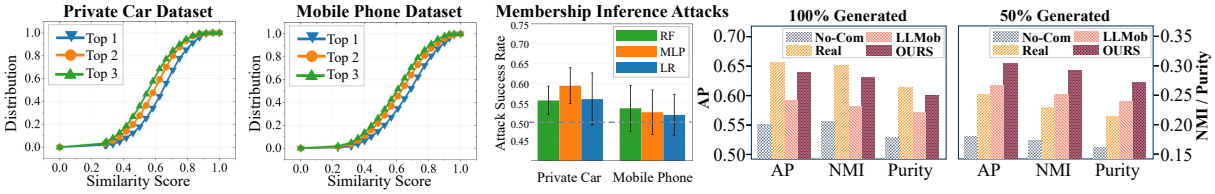


Figure 10: Uniqueness testing on two datasets (left). Membership inference attack accuracy (Middle). Downstream prediction on the private car dataset under the 100% and 50% settings; higher scores indicate better performance (Right).

4.11 Privacy Analysis

We assess privacy through uniqueness testing and membership inference attacks (MIA) [5, 45]. Using normalized Levenshtein similarity [16], the left two panels of Fig. 10 show moderate similarity between generated and training trajectories on both datasets, suggesting limited direct reproduction. For MIA, the attacker uses four Levenshtein-based features without access to training records, model parameters, prompts, or identities. We evaluate 134 members and 134 non-members across five stratified, user-disjoint splits using RF [4], MLP [32], and LR [7]. As shown in the middle panel of Fig. 10, attack accuracies are modestly above the 50% random-guessing baseline. These results suggest no evidence of severe privacy exposure under the evaluated attack protocol.

4.12 Downstream Task Analysis

We evaluate synthetic mobility using CDGP [13] for future user-location interaction prediction (AP) and community recovery (NMI and Purity). We compare CDGP trained on real data (**Real**), without community information (**No-Com**), and on data generated by **LLMob** or **MOTION**, using the same held-out real test set. As shown in the right two panels of Fig. 10, MOTION outperforms LLMob and approaches Real when using 100% generated training data. In the 50% setting, LLMob and MOTION augment half the real data with an equal amount of synthetic data, while Real and No-Com use only the real half, demonstrating its utility for real-world interaction prediction and community modeling.

5 RELATED WORK

Human mobility generation has evolved from mechanistic models to deep generators and LLM-based agents. Early rule-based methods such as DRTISA [28] and d-EPR [27] encode routines, exploration, and preferential return but rely on simplified behavioral assumptions. Deep models such as SeqGAN [42] and MoveSim [11] learn user preferences and spatiotemporal transitions through adversarial generation. Large language models (LLMs) provide strong semantic understanding and contextual reasoning for mobility modeling. LLMob [36] verbalizes trajectory histories and contextual information for autoregressive next-location inference, whereas LLM-COD [41] conditions destination generation on departure time, origin, and regional POI semantics. An LLM-based agent further equips an LLM with memory, planning, tool use, and iterative feedback, enabling interpretable multi-step decisions and the integration of heterogeneous urban knowledge. Ju et al. [14] developed TrajLLM to hierarchically generate personas, activities, and destinations; Feng et al. [9] introduced AgentMove, which combines spatiotemporal memory, world knowledge, and collective patterns for zero-shot next-location reasoning. Future research may use multi-agent collaboration to simulate interactions among individuals and uncover deeper relational structures underlying human mobility [17, 29].

6 CONCLUSION

In this paper, we presented MOTION, a prior-guided Multi-agent framework for cOmmuniTy-consIstent human mObility geNeration. MOTION extracts community-conditioned interaction priors, employs user agents and location agents to propose complementary graph updates, and uses a manager agent for coordination and community-consistency reflection. Experiments on two real-world datasets demonstrate improvements across individual, population, and community levels, while further analyses confirm cross-city prior recombination, privacy preservation, and downstream utility.

REFERENCES

[1] Hong-Kyun Bae, Yebeen Kim, Hyunjoon Kim, and Sang-Wook Kim. 2024. Negative sampling in next-POI recommendations: Observation, approach, and evaluation. In *Proceedings of the ACM Web Conference 2024*. 3888–3899.

[2] Aleix Bassolas, Hugo Barbosa-Filho, Brian Dickinson, Xerxes Dotiwalla, Paul Eastham, Riccardo Gallotti, Gourab Ghoshal, Bryant Gipson, Surendra A Hazarie, Henry Kautz, et al. 2019. Hierarchical organization of urban mobility and its connection with city livability. *Nature communications* 10, 1 (2019), 4817.

[3] Aleksandar Bojchevski, Oleksandr Shchur, Daniel Zügner, and Stephan Günnemann. 2018. Netgan: Generating graphs via random walks. In *International conference on machine learning*. PMLR, 610–619.

[4] Leo Breiman. 2001. Random forests. *Machine learning* 45, 1 (2001), 5–32.

[5] Zhongcai Cao, Kang Liu, Xin Jin, Li Ning, Ling Yin, and Feng Lu. 2025. Stage: a spatiotemporal-knowledge enhanced multi-task generative adversarial network (gan) for trajectory generation. *International Journal of Geographical Information Science* 39, 5 (2025), 1100–1127.

[6] Chen Chu, Hengcai Zhang, Peixiao Wang, and Feng Lu. 2024. Simulating human mobility with a trajectory generation framework based on diffusion model. *International Journal of Geographical Information Science* 38, 5 (2024), 847–878.

[7] David R Cox. 1958. The regression analysis of binary sequences. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 20, 2 (1958), 215–232.

[8] Bangchao Deng, Xin Jing, Tianyue Yang, Bingqing Qu, Dingqi Yang, and Philippe Cudre-Mauroux. 2025. Revisiting synthetic human trajectories: Imitative generation and benchmarks beyond datasaurus. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 1*. 201–212.

[9] Jie Feng, Yuwei Du, Jie Zhao, and Yong Li. 2025. Agentmove: A large language model based agentic framework for zero-shot next location prediction. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. 1322–1338.

[10] Jie Feng, Yong Li, Chao Zhang, Funing Sun, Fanchao Meng, Ang Guo, and Depeng Jin. 2018. Deepmove: Predicting human mobility with attentional recurrent networks. In *Proceedings of the 2018 world wide web conference*. 1459–1468.

[11] Jie Feng, Zeyu Yang, Fengli Xu, Haisu Yu, Mudan Wang, and Yong Li. 2020. Learning to simulate human mobility. In *Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining*. 3426–3433.

[12] Letian Gong, Yan Lin, Xinyue Zhang, Yiwen Lu, Xuedi Han, Yichen Liu, Shengnan Guo, Youfang Lin, and Huaiyu Wan. 2024. Mobility-llm: Learning visiting intentions and travel preference from human mobility data with large language models. *Advances in Neural Information Processing Systems* 37 (2024), 36185–36217.

[13] Shuo Ji, Xiaodong Lu, Mingzhe Liu, Leilei Sun, Chuanren Liu, Bowen Du, and Hui Xiong. 2023. Community-based dynamic graph learning for popularity prediction. In *Proceedings of the 29th ACM SIGKDD conference on knowledge discovery and data mining*. 930–940.

[14] Chenlu Ju, Jiaxin Liu, Shobhit Sinha, Hao Xue, and Flora Salim. 2025. Trajllm: A modular llm-enhanced agent-based framework for realistic human trajectory simulation. In *Companion Proceedings of the ACM on Web Conference 2025*. 2847–2850.

[15] Xiaochong Lan, Jie Feng, Yizhou Sun, Chen Gao, Jiahuan Lei, Xinleishi Xinleishi, Hengliang Luo, and Yong Li. 2025. Open-set living need prediction with large language models. In *Findings of the Association for Computational Linguistics: ACL 2025*. 5454–5472.

[16] Vladimir I Levenshtein et al. 1966. Binary codes capable of correcting deletions, insertions, and reversals. In *Soviet physics doklady*, Vol. 10. Soviet Union, 707–710.

[17] Jiachen Li, Fan Yang, Masayoshi Tomizuka, and Chiho Choi. 2020. Evolvegraph: Multi-agent trajectory prediction with dynamic relational reasoning. *Advances in neural information processing systems* 33 (2020), 19783–19794.

[18] Tong Li and Yong Li. 2023. Artificial intelligence for reducing the carbon emissions of 5G networks in China. , 1522–1523 pages.

[19] Xuchuan Li, Fei Huang, Jianrong Lv, Zhixiong Xiao, Guolong Li, and Yang Yue. 2024. Be more real: Travel diary generation using llm agents and individual profiles. *arXiv preprint arXiv:2407.18932* (2024).

[20] Yuxuan Liang, Kun Ouyang, Lin Jing, Sijie Ruan, Ye Liu, Junbo Zhang, David S Rosenblum, and Yu Zheng. 2019. Urbanfm: Inferring fine-grained urban flows. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*. 3132–3142.

[21] Yuxuan Liang, Kun Ouyang, Junkai Sun, Yiwei Wang, Junbo Zhang, Yu Zheng, David Rosenblum, and Roger Zimmermann. 2021. Fine-grained urban flow prediction. In *Proceedings of the Web Conference 2021*. 1833–1845.

[22] Bo Liu, Tong Li, Zhu Xiao, Wenjie Chen, Yongqiang Hei, Jingtao Ding, and Guo Chen. 2026. A multi-granularity self-guiding graph diffusion model for predicting private car activity. *IEEE Transactions on Intelligent Transportation Systems* (2026).

[23] Hao Liu, Ying Li, Yanjie Fu, Huaibo Mei, Jingbo Zhou, Xu Ma, and Hui Xiong. 2020. Polestar: An intelligent, efficient and national-wide public transportation routing engine. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 2321–2329.

[24] Jing Long, Tong Chen, Guanhua Ye, Kai Zheng, Quoc Viet Hung Nguyen, and Hongzhi Yin. 2024. Physical trajectory inference attack and defense in decentralized poi recommendation. In *Proceedings of the ACM Web Conference 2024*. 3379–3387.

[25] Yingtao Luo, Qiang Liu, and Zhaocheng Liu. 2021. Stan: Spatio-temporal attention network for next location recommendation. In *Proceedings of the web conference 2021*. 2177–2185.

[26] Giovanni Mauro, Massimiliano Luca, Antonio Longa, Bruno Lepri, and Luca Pappalardo. 2022. Generating mobility networks with generative adversarial networks. *EPJ data science* 11, 1 (2022), 58.

[27] Luca Pappalardo, Salvatore Rinzivillo, and Filippo Simini. 2016. Human mobility modelling: exploration and preferential return meet the gravity model. *Procedia Computer Science* 83 (2016), 934–939.

[28] Luca Pappalardo and Filippo Simini. 2018. Data-driven generation of spatio-temporal routines in human mobility. *Data Mining and Knowledge Discovery* 32, 3 (2018), 787–829.

[29] Joon Sung Park, Joseph O'Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th annual acm symposium on user interface software and technology*. 1–22.

[30] Roberto Pellungrini, Luca Pappalardo, Francesca Pratesi, and Anna Monreale. 2017. A data mining approach to assess privacy risk in human mobility data. *ACM Transactions on Intelligent Systems and Technology (TIST)* 9, 3 (2017), 1–27.

[31] Can Rong, Jingtao Ding, and Yong Li. 2024. An interdisciplinary survey on origin-destination flows modeling: Theory and techniques. *Comput. Surveys* 57, 1 (2024), 1–49.

[32] Frank Rosenblatt. 1958. The perceptron: a probabilistic model for information storage and organization in the brain. *Psychological review* 65, 6 (1958), 386.

[33] Yossi Rubner, Carlo Tomasi, and Leonidas J Guibas. 1998. A metric for distributions with applications to image databases. In *Sixth international conference on computer vision (IEEE Cat. No. 98CH36271)*. IEEE, 59–66.

[34] Chenyang Shao, Fengli Xu, Bingbing Fan, Jingtao Ding, Yuan Yuan, Meng Wang, and Yong Li. 2024. Chain-of-planned-behaviour workflow elicits few-shot mobility generation in llms. *arXiv preprint arXiv:2402.09836* (2024).

[35] Vincent A Traag, Ludo Waltman, and Nees Jan Van Eck. 2019. From Louvain to Leiden: guaranteeing well-connected communities. *Scientific reports* 9, 1 (2019), 5233.

[36] Jiawei Wang, Renhe Jiang, Chuang Yang, Zengqing Wu, Makoto Onizuka, Ryosuke Shibasaki, Noboru Koshizuka, and Chuan Xiao. 2024. Large language models as urban residents: An llm agent framework for personal mobility generation. *Advances in Neural Information Processing Systems* 37 (2024), 124547–124574.

[37] Yu Wang, Tongya Zheng, Yuxuan Liang, Shunyu Liu, and Mingli Song. 2024. Cola: Cross-city mobility transformer for human trajectory simulation. In *Proceedings of the ACM Web Conference 2024*. 3509–3520.

[38] Yang Xu, Gao Cong, Lei Zhu, and Lizhen Cui. 2024. MMPOI: A multi-modal content-aware framework for POI recommendations. In *Proceedings of the ACM web conference 2024*. 3454–3463.

[39] Yanyan Xu, Luis E Olmos, David Mateo, Alberto Hernando, Xiaokang Yang, and Marta C González. 2023. Urban dynamics through the lens of human mobility. *Nature computational science* 3, 7 (2023), 611–620.

[40] Jiaxuan You, Rex Ying, Xiang Ren, William Hamilton, and Jure Leskovec. 2018. Graphrnn: Generating realistic graphs with deep auto-regressive models. In *International conference on machine learning*. PMLR, 5708–5717.

[41] Chenyang Yu, Xinpeng Xie, Yan Huang, and Chenxi Qiu. 2024. Harnessing llms for cross-city od flow prediction. In *Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems*. 384–395.

[42] Lantao Yu, Weinan Zhang, Jun Wang, and Yong Yu. 2017. Seqgan: Sequence generative adversarial nets with policy gradient. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 31.

[43] Jing Yuan, Yu Zheng, and Xing Xie. 2012. Discovering regions of different functions in a city using human mobility and POIs. In *Proceedings of the 18th ACM SIGKDD international conference on Knowledge discovery and data mining*. 186–194.

[44] Yuan Yuan, Jingtao Ding, Depeng Jin, and Yong Li. 2025. Learning the complexity of urban mobility with deep generative network. *PNAS nexus* 4, 5 (2025), pgaf081.

[45] Yuan Yuan, Jingtao Ding, Huandong Wang, and Depeng Jin. 2024. Generating daily activities with need dynamics. *ACM Transactions on Intelligent Systems and Technology* 15, 2 (2024), 1–28.

[46] Shiyuan Zhang, Tong Li, Shuodi Hui, Guangyu Li, Yanping Liang, Li Yu, Depeng Jin, and Yong Li. 2023. Deep transfer learning for city-scale cellular traffic generation through urban knowledge graph. In *Proceedings of the 29th ACM SIGKDD conference on knowledge discovery and data mining*. 4842–4851.

[47] Yuheng Zhang, Yuan Yuan, Jingtao Ding, Jian Yuan, and Yong Li. 2025. Noise matters: Diffusion model-based urban mobility generation with collaborative noise priors. In *Proceedings of the ACM on Web Conference 2025*. 5352–5363.

SUPPLEMENTARY MATERIAL

Table 6: Statistics of the private car and mobile phone datasets.

Dataset	Period	# Locations	# Users	# POIs	# Records	Avg. Moves per User	Avg. Daily Moves per User
Private car dataset	2 weeks	500	2,342	717,461	678,341	8.79	0.88
Mobile phone dataset	2 weeks	500	5,352	679,413	1,235,619	18.57	1.28

A DATASET STATISTICS AND POI PROFILES

Both datasets span two weeks, covering 2,342 private vehicles and 5,352 mobile users (Table 6). Records contain only de-identified IDs, timestamps, and region-level locations. Data were collected with explicit authorization from users and providers and anonymized before we accessed them. Besides, we aggregate an open-source POI database into 14 top-level categories, with finer-grained subcategories underneath. Each region's category proportions form a profile for semantic mobility planning, destination retrieval, and location-side reasoning.

B BASELINE DESCRIPTIONS

MoveSim [11] conditions adversarial trajectory generation on user identity and historical movements to learn destination preferences and spatial transitions.

DeepMobility [44] couples individual mobility and population flows through adversarial imitation learning, bottom-up interaction modeling, and top-down feedback.

CoDiffMob [47] uses collaborative diffusion noise priors derived from population flows and individual preferences to generate realistic mobility trajectories.

LLMob [36] generates personalized trajectories autoregressively from LLM instructions encoding historical trajectories, time, locations, and semantic context.

LLM-COD [41] predicts destination semantics from departure, time, and POI context, then matches candidate regions through conditional language generation.

TrajLLM [14] hierarchically constructs user personas, selects activities, and determines destinations through an LLM-enhanced agent framework for daily trajectory simulation.

AgentMove [9] combines individual spatiotemporal memory, urban world knowledge, and shared population patterns in an agentic framework for next-location prediction.

C DETAILED EXPERIMENTAL SETTINGS

MOTION uses Python/LangGraph and GPT-4o (temperature 0.2, seed 42), with hourly steps over 500 regions, `top_k=active_m=10`, and at most five reflection rounds. Users are split 8:2 for training/testing; priors use training data preceding the final day. We generate 24 steps from test users' midnight locations, sharing initialization with compatible baselines. Graph construction uses 3-km spatial decay, a 10-km cutoff, and the top 20% of location edges by combined spatial-flow weight. Leiden uses resolution 0.2, 20 initial and five warm-start iterations, and temporal smoothing 0.4. Retrieval uses population exponent 1.0, POI multiplier 2.0, minimum distance 0.1 km, and exploration probability 0.05. Location activation prioritizes incoming activity; the co-mobility threshold is 0.6. Reflection compares graphs with training priors and revises only affected users/locations within the current step.

D DETAILS OF EVALUATION METRICS

We compare real and generated trajectories using three metric groups [11, 26]. Let d denote geographic distance and T the number of time steps. We measure distributional differences using the square root of the base-2 Jensen–Shannon divergence:

$$\mathrm{JSD}(P,Q) = \sqrt{\frac{\mathrm{KL}_2(P\|M) + \mathrm{KL}_2(Q\|M)}{2}},$$

where $M = (P+Q)/2$ and JSD lies in $[0,1]$.

D.1 Individual Trajectory Statistics

We report JSD between the real and generated distributions of each statistic [11, 28]. **Distance:** Mean geographic displacement between consecutive time steps in a trajectory. **Radius:** Geographic distance of visited locations from the trajectory center, capturing the spatial extent of movement. **LocNum:** Number of distinct locations visited.

D.2 Population and Flow Distributions

We assess spatial population and flow patterns [31] using Earth Mover's Distance [33]:

$$\mathrm{EMD}_d(P,Q) = \min_{\pi\in\Pi(P,Q)} \sum_{x,y} \pi(x,y)d(x,y),$$

where $\Pi(P,Q)$ contains transport plans with marginals P and Q.

Pop-EMD. We average $\mathrm{EMD}_d(P_t^{\mathrm{pop}}, \widehat{P}_t^{\mathrm{pop}})$ over time, where P_t^{pop} and $\widehat{P}_t^{\mathrm{pop}}$ are real and generated regional population distributions.

Flow-EMD. We average EMD between real and generated origin–destination event distributions over time, including stays and using ground distance d_F. For events $e = (o,r)$ and $e' = (o',r')$, the ground distance is $d_F(e,e') = [d(o,o') + d(r,r')]/2$.

G-Rank. On the shared set $\mathcal{L}_{100}$ of the 100 most visited real locations, we compare normalized visit counts using $\mathrm{JSD}(\mathbf{p},\widehat{\mathbf{p}})$, where $p_\ell = n_\ell / \sum_{j\in\mathcal{L}_{100}} n_j$ and $\widehat{p}_\ell$ is defined from generated counts.

D.3 Community Structural Consistency

These metrics assess community organization [2, 13] and graph structure [3, 26].

Motif. We compute JSD between normalized occurrence distributions of the five non-stay motifs in Table 7.

Comm-Size. Comm-Size measures community size consistency across sliding windows. For each window $w \in \mathcal{W}$, Leiden detects communities from aggregated user-location graphs. Given partition $\mathcal{C}_w$, the size distribution is:

$$p_w^{\mathcal{C}}(b) = \frac{1}{|\mathcal{C}_w|} \sum_{C\in\mathcal{C}_w} \mathbb{I}\left[\frac{|C|}{|\mathcal{V}_w|} \in b\right], \quad b \in \mathcal{B},$$

where $\mathcal{V}_w$ is the node set and $\mathcal{B}$ denotes shared bins. We compute:

$$\text{Comm-Size} = \frac{1}{|\mathcal{W}|} \sum_{w \in \mathcal{W}} \text{JSD}(p_w^C, \hat{p}_w^C).$$

Comm-Mem. Comm-Mem evaluates community membership consistency. We aggregate visits over 23 hourly steps into bipartite graphs with unit-weight user-location edges and no static backbone. Leiden with modularity (resolution 1.0) independently detects communities. Let A_k^L and B_k^L denote shared location labels from real and generated graphs under seeds k and $100 + k$. We compute:

$$\text{Comm-Mem} = \frac{1}{5} \sum_{k=0}^{4} \frac{2I(A_k^L; B_k^L)}{H(A_k^L) + H(B_k^L)},$$

where I and H denote mutual information and entropy. Higher Comm-Mem indicates better membership consistency.

Comm-Trans. Comm-Trans measures community evolution using overlapping 8-hour windows with a 1-hour stride. Leiden detects communities from aggregated user-location graphs without static edges. Let $c_w(\ell_i)$ be the community label of location ℓ_i. We define $S_w(i,j) = \mathbb{I}[c_w(\ell_i) = c_w(\ell_j)]$, $\Delta_w(i,j) = S_{w+1}(i,j) - S_w(i,j)$, where $\Delta_w \in -1, 0, 1$ indicates separation, no change, or convergence. Let $\mathcal{T}^R$ and $\mathcal{T}^G$ be the real and generated nonzero transitions $(w, i, j, \Delta_w(i,j))$. We compute:

$$\text{Comm-Trans} = \frac{2|\mathcal{T}^R \cap \mathcal{T}^G|}{|\mathcal{T}^R| + |\mathcal{T}^G|}.$$

E DETAILS OF MOTIF MINING

We extract six mobility motifs through automatic metagraph-based prior mining (Table 7). Distinct letters denote distinct locations; repeated letters indicate revisits to the same location instance. Each movement carries source and destination POI semantics and one of seven distance bins: 0–1, 1–2, 2–3, 3–5, 5–8, 8–12, and > 12 km. For example, MOVE_AB may represent Commercial & Residential → Companies & Enterprises over 3–5 km, linking a structural pattern to semantic mobility planning. STAY denotes a continuous stay and has no distance bin.

Table 7: Mobility motifs mined from location sequences.

Motif	Sequence	Meaning
STAY	A	No movement
MOVE_AB	$A \to B$	One-way move
ROUND_ABA	$A \to B \to A$	Return to source
CHAIN_ABC	$A \to B \to C$	Three distinct locations
RETURN_ABCB	$A \to B \to C \to B$	Intermediate return
FULL_ABCBA	$A \to B \to C \to B \to A$	Nested return

F ROBUSTNESS TO SPATIAL BACKBONE

To examine whether community consistency depends on static backbone information or specific user-location interactions, we evaluate three perturbations: randomizing location-location flow weights (**Randomized**), removing location-location edges (**UL-only**), and degree-preserving random rewiring of user-location edges (**UL-shuffled**). We aggregate interactions over 23 hourly steps for community detection. As shown in Table 8, MOTION maintains its advantage without the original spatial backbone. The UL-shuffled control further examines whether community agreement reflects co-visitation structure beyond node-level visit statistics.

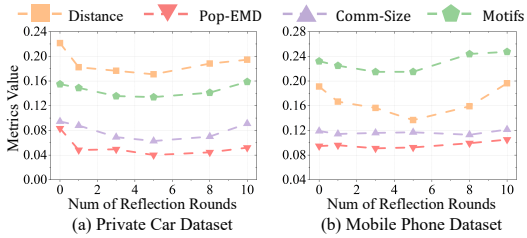
Table 8: Community consistency under backbone perturbations (NMI↑).

Setting	CoDiffMob	DeepMobility	TrajLLM	LLMob	MOTION
Randomized	0.73 ± 0.04	0.73 ± 0.03	0.70 ± 0.01	0.65 ± 0.03	**0.76** ± 0.02
UL-only	0.55 ± 0.03	0.54 ± 0.02	0.51 ± 0.01	0.40 ± 0.02	**0.58** ± 0.02
UL-shuffled	0.42 ± 0.03	0.43 ± 0.02	0.40 ± 0.04	0.28 ± 0.02	**0.45** ± 0.03

Table 9: Comparison with baselines supplied with reflection information ("+") on the private car dataset.

Method	Dist.↓	LocNum↓	P-EMD↓	C-Size↓	C-Mem↑	C-Trans.↑
CoDiffMob+	0.2097	0.2523	0.0422	0.0663	0.5490	0.1524
LLM-COD+	0.2173	0.1963	0.0486	0.0736	0.5461	0.1511
MOTION	**0.1709**	**0.1258**	**0.0402**	**0.0629**	**0.5766**	**0.1990**

Dist.=Distance; P-EMD=Pop-EMD; C-Size/Mem/Trans=Comm-Size/Mem/Trans.



(a) Private Car Dataset (b) Mobile Phone Dataset

Figure 11: Impact of the maximum number of within-step reflection rounds. Lower values indicate better performance.

G REFLECTION ANALYSIS

Fig. 11 shows that performance generally improves over the first five reflection rounds, with inconsistent gains thereafter; we therefore use five rounds. We also add reflection information to CoDiffMob and LLM-COD ("+") on the private car dataset. Specifically, we compute deviations from priors, using them to guide deterministic post-generation trajectory corrections in CoDiffMob+ and to construct deviation-specific prompts for trajectory refinement in LLM-COD+. MOTION leads all metrics in Table 9, suggesting benefits from coordinated multi-agent reasoning beyond access to reflection information.

H ADDITIONAL RECOMBINATION RESULTS

Fig. 12 shows results of transferring source-dataset mobility motifs while retaining target-dataset interaction priors. Distance and Radius show larger transfer gaps, particularly from private car to mobile phone data, reflecting differences in travel modes and spatial scales. Pop-EMD, Flow-EMD, Comm-Size, and Motif are more stable: for this transfer, Flow-EMD and Comm-Size increase slightly. Thus, target interaction priors help preserve collective structure despite changes in individual mobility scale.

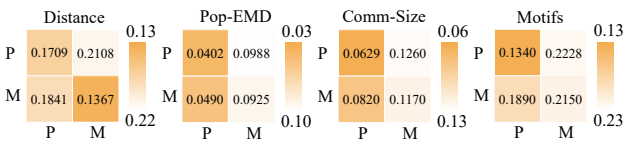


Figure 12: Cross-dataset prior recombination between the private car dataset (P) and mobile phone dataset (M). Rows/Columns indicate the source/target dataset.

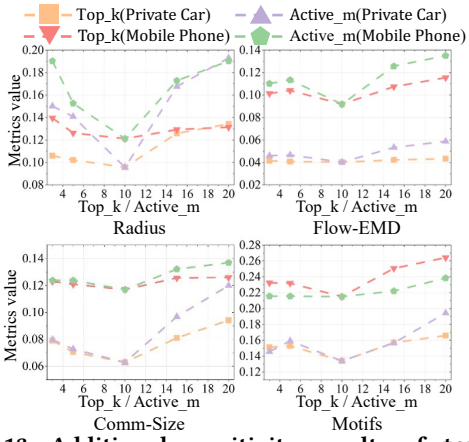


Figure 13: Additional sensitivity results of `top_k` and `active_m` on the private car and mobile phone datasets.

I ADDITIONAL HYPERPARAMETER RESULTS

We vary `top_k` (destination candidates per user) and `active_m` (locations sampled per local set) separately. Fig. 13 complements the main-text Comm-Mem and Pop-EMD results: most curves reach their minima near 10 on both datasets. Smaller values limit destination and community evidence; values of 15 or 20 introduce less relevant context, degrading Comm-Size and Motif while increasing cost. These results support setting both parameters to 10.

J CONTINUOUS MULTI-DAY GENERATION

Daily resetting outperforms uninterrupted generation over two to four days (Table 10). Without resetting, Pop-EMD remains at 0.0444–0.0452, while Comm-Mem falls from 0.5473 to 0.5304, indicating gradual membership drift despite stable population distributions.

Table 10: Multi-day generation with daily resetting (Reset) or uninterrupted generation (No Reset).

Days	Pop-EMD↓		Comm-Mem↑	
	Reset	No Reset	Reset	No Reset
2	**0.0417**	0.0451	**0.5642**	0.5473
3	**0.0415**	0.0444	**0.5779**	0.5371
4	**0.0348**	0.0452	**0.5803**	0.5304

K AGENT PROMPT TEMPLATES

Figs. 16–17 provide prompt templates for agent reasonings.

User Agent Reasoning Prompt Example

Instruction: You are simulating a mobility agent (<USER_LABEL>) in <CITY> at hour <CURRENT_HOUR>. Your agent background is that you are typically active in <COMMUNITY_PROFILE> areas. Your current action is departing from <FROM_POI>, heading toward a <TO_POI> area at approximately <DIST_LABEL>. You are given a list of candidate destinations <CANDIDATE_LIST>, sorted by gravity score, where flow_out represents each candidate's total historical outbound traffic. Choose the best destination by holistically balancing three factors: a POI match combined with decent gravity and flow is better than the highest gravity alone; a zero-flow_out location should only be chosen as a last resort when no active alternative exists; and you should not fixate on a single dimension, but consider each candidate's combined profile. Reply with a JSON object only, specifying the next location id chosen from the candidate list above and a reason of no more than 25 words.
Instructions (continued): <REFLECTION_SIGNAL> is appended verbatim after the base prompt above, with no additional framing.

Figure 14: Prompt template for user agent reasoing.

Location Agent Reasoning Prompt Example

Instruction: You are location <LOC_ID> (<POI_TYPES>) in <CITY> at hour <CURRENT_HOUR>. Your role is to decide which nearby <USER_LABEL> are likely to move to your location this hour. Your location profile is as follows: POI types <POI_TYPES> (ranked by POI density), current population status <POP_STATUS>, a simulated inbound flow of <SIM_FLOW_IN> <USER_LABEL> currently waiting at source locations, and a historical inbound flow of approximately <HIST_FLOW_IN> <USER_LABEL> per hour at this time of day, which serves as a rough ceiling on how many you can realistically attract. Your main historical source locations, ranked by importance, are <SOURCE_LOCATIONS>. The <USER_LABEL> currently at those source locations in the previous timestep are <NEARBY_USERS>, and the co-mobility groups active in this neighborhood — pairs of <USER_LABEL> who tend to travel together—are <CO_MOBILITY_GROUPS>. When deciding whom to attract, respect the population signal above for whether to attract at all this hour; among those you do consider, prioritize <USER_LABEL> whose planned destination POI matches your own POI types, then <USER_LABEL> coming from higher-ranked source locations with more people currently present, and try to keep co-mobile pairs together when both are moving. <USER_LABEL> currently planning to stay are equally valid attraction targets, not just those already planning to move. Please analyze the situation and return your decision as a JSON object, listing only user ids drawn from the nearby <USER_LABEL> list above together with a brief reasoning of no more than 25 words.
Instructions (continued): This round, a reflection correction from the previous round is provided as <REFLECTION_SIGNAL>. Note that this is a correction to what you already decided last round, not merely additional context.

Figure 15: Prompt template for location agent reasoning.

Manager Agent Coordination Prompt Example

Instruction: You are coordinating mobility decisions for <USER_LABEL> in <CITY> at hour <CURRENT_HOUR>. Two independently-reflected mobility views — a user-intent view and a location-attraction view — each propose a decision for every record in <CONFLICT_RECORDS>; some records agree between the two views and some conflict. For each record, decide the single final outcome by weighing both views' stated reasons against two real-data signals: this hour's expected total move count, given as <MOVE_COUNT_RANGE>, and each candidate destination's own deviation from the real historical population and flow, shown inline as <VS_REAL_HINTS> next to each candidate. If the total MOVE resolutions would clearly overshoot or undershoot the expected range, favor whichever candidate per record brings the total closer to that range, subject to the rule above; otherwise, prefer the candidate closer to the real historical headcount, avoiding over-populated destinations when an alternative exists. Reply with a JSON object only, listing for every record the user id, the resolved destination location id (must be one of that record's listed candidates), the resulting action (MOVE when the resolved destination differs from the origin, otherwise STAY), and a reason of no more than 20 words.
Instructions (continued): This round, a reflection correction from the previous round is provided as <REFLECTION_SIGNAL>. Note that this is a correction to what you already decided last round, not merely additional context.

Figure 16: Prompt template for manager agent coordination.

Manage Agent Reflection Prompt Example

Instruction: You are a mobility simulation quality analyst for <CITY>. At each hour, compare the generated user-location graph with historical community priors. You are given community deviations <COMMUNITY_DIFFS>, population deviations <NODE_DIFFS>, flow deviations <EDGE_DIFFS>, affected users' plans <USER_DAILY_PLANS>, and movement context <HOUR_MOVE_STATUS>.
Community Deviation Diagnosis:
- COMMUNITY_SIZE_OVER / UNDER: A generated community is larger or smaller than its historical reference.
- COMMUNITY_MEM_MISMATCH: A location's detected community membership differs from its aligned historical reference, indicating inconsistent interaction patterns.
- ……

Targeted Reflection:
Identify responsible users and locations, then generate targeted corrections:
- User signals: Revise movement actions, destinations, or travel ranges while respecting individual mobility intentions.
- Location signals: Adjust attraction, preferred source locations, or candidate users to improve community consistency.

Do not directly modify community assignments or force historical trajectories. Instead, improve community structures through revised user-location interactions.
Generate at most <MAX_USER_SIGNALS> user signals and <MAX_LOCATION_SIGNALS> location signals. Reference only IDs provided in the current deviation records. Return a JSON object containing user_signals and location_signals following <SIGNAL_JSON_TEMPLATE>.

Figure 17: Prompt template for manager agent reflection.